MIL-M-14927B (AR)
3 NOVEMBER 1977
SUPERSEDING
MIL-M-14927A (PA)
3 June 1974

MILITARY SPECIFICATION

MOTOR, ROCKET, LAUNCH, M114
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the loading, assembling and packing of one type of rocket motor designated as Motor, Rocket, Launch, M114.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents.-The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic

FSC: 1336

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

9225166	-	Motor Launch Rocket (TOW) M114
9231698	-	Drum, Metal, Shipping, for Rocket Motor, Launch (TOW) M114
9298138	-	Packing and Marking for Drum, Fiber, Shipping for Rocket Motor, Launch (TOW) M114

INSPECTION EQUIPMENT DRAWINGS

9225710	-	Support, Propellant, Test
9225711	-	Metal Parts Assembly, TOW Launch Motor, M114 Test
9225714	-	Loading and Firing Procedures

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Rocket Motor.-The motor shall comply with all the requirements specified on drawing (dwg.) 9225166.

3.3 Propellant Grain Integrity

3.3.1 External.-The grain shall be free of surface flaws.

3.3.2 Internal.-The pin-stick assembly, dwg. 9225164, shall be free of discontinuities, cracks, voids and inclusions. The assembly shall not exhibit a reamed surface beyond the tapered portion of the jacketed pin.

3.4 Ballistic Performance.-The motor shall comply with the requirements as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

TABLE I - BALLISTIC REQUIREMENTS

	-25°F		+125°F	
	Min.	Max.	Min.	Max
Thrust, lbf.	NR	NR	-	15000
Pressure, psig.	NR	NR	-	13000
Impulse, lbf,-sec	257	-	NR	NR
Action Time, msec	-	67	NR	NR
Ignition Delay, msec	-	27	-	27

NR = No Requirement
msec = milliseconds

3.5 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.-Parts shall be clean, free of sharp corners, burrs, chips and other foreign material. Assemblies shall be free of excess solder and adhesive. Exterior surface coatings shall be continuous except for light scratches not exposing base material. Markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Insert, Suspension Pin	9218515	10
Jacketed Suspension Pin	9218516	10
Grain, Propellant	9225163	10
Syrup, Dipping	9268941	250 cubic Centimeters
Pin-Stick Assembly	9225164	10
Motor, Launch, Rocket (TOW) M114	9225166	15

4.3.2 Inspections to be Performed.-See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection.-See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Motor, Launch, Rocket (TOW) M114				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Grain, Propellant (Dwg. 9225163)				
	Examination for Defects	10	-	3.2	4.4.2.1
	Insert, Suspension Pin (Dwg. 9218515)				
	Examination for Defects	10	-	3.2	4.4.2.2
	Jacketed Suspension Pin (Dwg. 9218516)				
	Examination for Defects	10	-	3.2	4.4.2.3
	Syrup, Dipping (Dwg. 9268941)				
	Examination for Defects	250CC	-	3.2	4.4.2.4
	Pin-Stick Assembly (Dwg. 9225164)				
	Examination for Defects	10	-	3.2	4.4.2.5
	External Grain Integrity	10	-	3.3.1	4.5.1.1
	Internal Grain Integrity	10	-	3.3.2	4.5.1.2
	Strength	10	-	3.2	4.5.2
	Motor Launch Rocket (TOW) M114 (Dwg. 9225166)				
	Examination for Defects	15	-	3.2	4.4.2.6
	Igniter Circuit Continuity	(a)+(b)	-	3.2	4.5.4
	Ballistic Performance Test at -25°F	(a)			
	Impulse, min.		0.65%	3.4	4.5.3
	Action Time, max.		0.65%	3.4	4.5.3
	Ignition Delay, max.		0.65%	3.4	4.5.3

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Continuation Sheet				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ballistic Performance Test at +125°F	(b)			
	Thrust, max.		0.65%	3.4	4.5.3
	Pressure, max.		0.65%	3.4	4.5.3
	Ignition Delay, max.		0.65%	3.4	4.5.3

NOTES: (a) and (b): Sample size and acceptance criteria in accordance with 4.4.3.3.

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded motors shall contain:

a. Inert parts from not more than one interfix lot number from one manufacturer.

b. Igniters from not more than one interfix lot number from one manufacturer.

c. Propellant grains from not more than one lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9225163
4.4.2.1	Grain, Propellant				NEXT HIGHER ASSEMBLY 9225164
CATEGORY	EXAMINATION OR TEST*	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Diameter of radial hole		0.40%	3.2	Gage
102	Presence of 360° shoulder		100%	3.2	Visual
Minor					
201	Taper improper		0.65%	3.2	Gage
202	Location of radial hole		0.65%	3.2	Gage
203	Length improper		0.65%	3.2	Gage
204	Straightness of outside diameter		0.65%	3.2	Gage
205	Inside diameter, min.		0.65%	3.2	Gage

NOTES: * Inspection to be performed at an ambient temperature of 70°F plus or minus 10°F.

SARPA-QA FORM 2567 FEB 74 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9218515
4.4.2.2	Insert, Suspension Pin				NEXT HIGHER ASSEMBLY 9218516
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Shank diameter of insert head		0.40%	3.2	Gage
102	Length of shank between flanges		0.40%	3.2	Gage
103	Outside diameter of flute		0.40%	3.2	Gage
Minor					
201	Outside diameter of flanged head		0.65%	3.2	Gage
202	Depth of head		0.65%	3.2	Gage
203	Perpendicularity of flange to outside diameter of flute		0.65%	3.2	Gage
204	Length of flute		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9218516
4.4.2.3	Jacketed Suspension Pin				NEXT HIGHER ASSEMBLY 9225164
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Major diameter of tapered jacket		0.40%	3.2	Gage
102	Minor diameter of tapered jacket		0.40%	3.2	Gage
103	Position of tapered minor diameter to insert shank diameter		0.40%	3.2	Gage
Minor					
201	Length of taper		0.65%	3.2	Gage
202	Length of jacket tip		0.65%	3.2	Gage
203	Diameter of jacket tip		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Syrup, Dipping				9268941
					NEXT HIGHER ASSEMBLY 9225163
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major 101	Viscosity	250cc		3.2	4.4.3.1
Minor	None defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Pin-Stick Assembly				9225164
					NEXT HIGHER ASSEMBLY
					9225166
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	External grain integrity		100%	3.3.1	4.5.1.1
102	Internal grain integrity		100%	3.3.2	4.5.1.2
103	Strength	(a)	1.50%	3.2	4.5.2
104	Length between flange and stick		0.40%	3.2	Gage
Minor	None defined.				

NOTES: (a) Sample size per MIL-STD-414, Inspection Level III.

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.6	Motor, Launch, Rocket (TOW) M114		9225166
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shorting wire improper		100%	3.2	Visual
2	Hydrostatic test stamp missing		100%	3.2	Visual
3(a)	Projection of support thread above top of nut		100%	3.2	Visual
Major					
101	Support nut torque		0.40%	3.2	Gage
102	Case-nozzle torque		0.40%	3.2	Gage
103	Weight		0.40%	3.2	Scale
104	Igniter damaged		0.40%	3.2	Visual
105	Spacer missing		0.40%	3.2	Visual
106	Igniter lead wires damaged		0.40%	3.2	Visual
107	Igniter circuit continuity		100%	3.2	4.5.4
	Ballistic Performance Test at -25°F	(b)			
108	Impulse, min.			3.4	4.5.3
109	Action Time, max.			3.4	4.5.3
110	Ignition Delay, max.			3.4	4.5.3
	Ballistic Performance Test at +125°F	(c)			
111	Thrust, max.		0.65%	3.4	4.5.3
112	Pressure, max.		0.65%	3.4	4.5.3
113	Ignition Delay, max.		0.65%	3.4	4.5.3

NOTES:
(a) This examination shall be performed after successful completion of the support nut torque test.
(b) Unique Double Sampling Plan in accordance with 4.4.3.2.
(c) The sample size and acceptance criteria shall be in accordance with MIL-STD-414, Section B, Part I, and Code Letter D.

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9225166
4.4.2.6	Continuation Sheet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking improper		0.65%	3.2	Visual
202	Protective finish improper		0.65%	3.2	Visual
203	Sealant missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Other Components					See Below
						NEXT HIGHER ASSEMBLY 9225166
CATEGORY	EXAMINATION OR TEST		NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Dwg. No.	Document Nomenclature				
	9225161	Spacer, Propellant				
	9225170	Cover, Protective				
	9260308	Splice, Parallel				
Critical	None defined.					
Major						
101	Component missing or improperly installed			0.40%	3.2	Visual
Minor						
201	Protective finish inadequate (if applicable)			0.65%	3.2	Visual
202	Evidence of poor workmanship			1.00%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9231698, 9298138 as applicable
4.4.2.8	Drum, Prior to Insertion of Cushion				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Motors improperly inserted in support		0.40%	3.2	Visual
102	Dessicant bags missing		0.40%	3.2	Visual
Minor	None defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9231698 9298138 as applicable
4.4.2.9	Drum, after Sealing of Barrier Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major 101	Heat seal improper		4.0%	3.2	Visual
Minor	None defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-14927B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9231698 9298138 as applicable
4.4.2.10	Drum, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	DOT marking missing or incorrect		0.40%	3.2	Visual
102	Contents loose when tilted		0.40%	3.2	Manual
103	Car seal missing or installed improperly		0.40%	3.2	Visual
Minor					
201	Marking missing, misleading or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Viscosity of Dipping Syrup.-(Dwg. 9268941) - When the prepared syrup is stored in unsealed containers, this test shall be performed once at the beginning of each shift, and when beginning a new syrup mix during the propellant dipping process. When the prepared syrup is stored in sealed containers, this test shall be performed after the formulation of the syrup mix and at two week intervals until used. Dipping syrup exhibiting a viscosity not as specified shall not be used.

4.4.3.2 Ballistic Performance Test at -25°F (Dwg. 9225166).- An initial sample of 15 motors shall be selected.

If $\bar{X}_{15} - 2.445s_{15} \geq$ LL and

$\bar{X}_{15} + 2.445s_{15} \leq$ UL accept the lot.

If $\bar{X}_{15} - 1.549s_{15} \leq$ LL or

$\bar{X}_{15} + 1.549s_{15} \geq$ UL reject the lot.

If no decision can be made, select an additional 30 motors.

If $\bar{X}_{45} - 2.065s_{45} \geq$ LL and

$\bar{X}_{45} + 2.065s_{45} \leq$ UL accept the lot.

The sample for this test shall be selected just prior to assembly of the motor lot. Randomly select from the following components which are to be assembled into the motor lot: pin stick assemblies (from one propellant lot), motor cases, nozzles and igniters.

4.4.3.3 Ballistic.-Acceptance for this test shall be in accordance with MIL-STD-414, Section B, Part I for single limits and Section B, Part II for double limits. The sample size and acceptance criteria for each temperature shall be:

Plus 125°F - Code Letter D.
Minus 25°F - Code Letter F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Method and Procedures

4.5.1 Propellant Grain Integrity

4.5.1.1 External.-The grain shall be visually examined in accordance with standards to be established prior to or during the First Article Sample examination.

4.5.1.2 Internal.-The pin-stick assembly shall be radiographically examined by directing a beam normal to the longitudinal axis. The assembly shall be radiographed twice, the second radiograph to be displaced 90 degrees from the first. Radiographic inspection procedures shall comply with MIL-STD-453. The penetrameter used shall be designed and constructed to insure a 2 percent sensitivity. If any assembly shows evidence of discontinuities, cracks, voids and inclusions or a reamed surface beyond the tapered portion of the jacketed pin, the assembly shall be classed defective.

4.5.2 Pin-Stick Assembly Strength.-The samples of Pin-Stick Assembly selected shall be conditioned to 70°F plus or minus 5°F and tested within 5 minutes at a load application rate of one (1) inch per minute applied to the propellant grain, jacketed suspension pin joint. The test shall be run to destruction and the results recorded. The lot shall be rejected if any failure is below 600 pounds. Only those test results which are directly associated with bond strength failures shall be included in the statistical analysis.

4.5.3 Ballistic Performance Tests.-The sample motors, Dwg. 9225712 shall consist of metal parts, Dwg. 9225711 with modified propellant support, Dwg. 9225710 and shall be loaded and fired as specified on Dwg. 9225714. The sample motors shall be conditioned at minus 25°, minus 5°F and plus 125°, plus 5°F. The motor conditioning time shall be 4 hours minimum. The time of motor removal from the conditioning chamber to the time of firing shall not exceed 5 minutes. Results shall be recorded and evaluated to determine compliance with the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

4.5.4 Igniter Circuit Continuity.-This test shall be performed on the launch motor assembly just prior to insertion of the assembly into the drum. The shorting wire shall be severed and the igniter leads shall be connected to a meter. The test current shall be limited to fifty (50) milliamperes, maximum. At the completion of this test, the shorting wire shall be reconnected with the parallel splice.

WARNING: THIS TEST MAY BE DANGEROUS AND SHOULD BE PERFORMED BEHIND A SUITABLE BARRICADE.

5. PACKAGING

5.1 Packing.-The launch motor assembly shall be packed in accordance with Dwg. 9231698 or Dwg. 9298138 as applicable.

5.2 Marking.-The containers shall be marked in accordance with Dwg. 9231698 or Dwg. 9298138 as applicable.

6. NOTES

6.1 Intended Use.-The Motor, Rocket, Launch, M114 covered by this specification is intended for use in the TOW Guided Missile, BGM-71A.

6.2 Ordering Data.-See MIL-A-48078.

6.3 Submission of Inspection Equipment for Design Approvals.-See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Submission of Test Data.-When required by incorporation of DD Form 1423 into the contract, all test data required in Section 4 of this specification shall be submitted in accordance with Data Item Description DI-R-1724. In addition to the normal distribution of records, one copy of all test data shall be forwarded to each of the following:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-M
DRDAR-LCU-D
Dover, New Jersey 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

6.5 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Definitions

6.6.1 Ballistic Definition.-Ballistic nomenclature shall be in accordance with MIL-STD-292.

6.6.2 Contractor.-Unless otherwise specified, the word "Contractor", used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

6.7 Changes from Previous Issue.-Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1336-A286

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1426 JAN 72

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:07Z
Check the source to verify that this is the current version before use.